Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 46: Approved Solutions

Finding Recurrences

Covers *MCS* Chapter 21

Status of this set

Ungraded practice. Solutions released with it.

Part I: How to Practice This Material

Setting up a recurrence

Ask: **what is the first choice, and what problem remains after it?**

- Disjoint first choices ⇒ **add** the resulting counts.
- Independent stages ⇒ **multiply**.
- Then state the boundary conditions — as many as the recurrence reaches back.

This is Lecture 31's counting with the answer written in terms of itself.

Three mistakes to expect

1. **Too few boundary conditions.** A recurrence reaching back two steps needs two.
2. **Overlapping cases.** The first choices must be mutually exclusive, or you double count.
3. **Treating a conjecture as proved.** Terms suggest; they do not establish.

Part II: Practice Problems

Section A — Writing the equation

1. A sequence satisfies $a_0 = 2$ and $a_n = 3a_{n-1}$.

(a) Compute a_0 through a_5.

(b) Conjecture a closed form.

(c) What would it take to prove your conjecture?

Approved solution

(a) $2, 6, 18, 54, 162, 486$.
(b) $a_n = 2 \cdot 3^n$.
(c) Either an induction proof — base $a_0 = 2 = 2 \cdot 3^0$, step $a_{k+1} = 3a_k = 3(2 \cdot 3^k) = 2 \cdot 3^{k+1}$ — or solving the recurrence directly, which is Wednesday's method.

Six matching terms is evidence, not proof. That lesson has not changed since Week 1.

2. How many bit strings of length n contain no two consecutive 1s? Set up a recurrence with boundary conditions, then compute the first six values.

Approved solution

Let $f(n)$ be the count. Split on the *first* bit:

- Starts with 0: the remaining $n-1$ bits are any valid string — $f(n-1)$ of them.
- Starts with 1: the next bit must be 0, and the remaining $n-2$ are free — $f(n-2)$ of them.

The cases are disjoint, so

$$f(1) = 2, \quad f(2) = 3, \qquad f(n) = f(n-1) + f(n-2) \quad (n \geq 3).$$

Values: $2, 3, 5, 8, 13, 21$.

Fibonacci again, with different boundary conditions. The recurrence captures the structure; the boundary conditions say which sequence you land on.
Two boundary conditions are required here because the recurrence reaches back two steps.

3. Into how many regions do n lines divide the plane, assuming no two are parallel and no three meet at a point? Set up a recurrence and find the first six values.

Approved solution

Let r_n be the number of regions, with $r_0 = 1$ (the whole plane).
Adding the nth line: it crosses each of the $n-1$ existing lines once, at $n-1$ distinct points, which cut it into n pieces. Each piece splits one existing region in two, adding n regions. So

$$r_0 = 1, \qquad r_n = r_{n-1} + n.$$

Values: $1, 2, 4, 7, 11, 16$.
Closed form: $r_n = 1 + \dfrac{n(n+1)}{2}$ — the Week 11 sum again.

The geometric reasoning is the hard part, and the algebra is routine once the equation is written. That is typical.

Section B — Reading a recurrence

4. For each, state how many boundary conditions are required and why.

(a) $f(n) = 5f(n-1)$

(b) $f(n) = f(n-1) + f(n-2)$

(c) $f(n) = f(n-1) + f(n-3)$

(d) $T(n) = 2T(n/2) + n$

Approved solution

(a) One. Each term needs only its predecessor.
(b) Two. Computing $f(2)$ needs both $f(1)$ and $f(0)$.
(c) Three. The recurrence reaches back three steps, so $f(0)$, $f(1)$, $f(2)$ must all be given.
(d) One, typically $T(1)$. The recursion halves until it reaches the base case, so only the smallest instance must be supplied.

The rule for the first three: as many boundary conditions as the furthest reach back. Part (d) is a different shape and follows a different rule — the reason it needs its own method on Friday.

Section C — Writing

5. In four to six complete sentences, explain why setting up a recurrence is a counting argument, using one of the problems above.

Approved solution

Model answer. A recurrence is produced by identifying a first choice that splits the problem into cases, counting how many ways each case can be completed, and expressing those counts in terms of smaller instances of the same problem. In the bit-string problem, the first bit is either 0 or 1, and those cases are mutually exclusive and exhaustive, so the Addition Principle says the total is the sum of the two counts. Choosing 0 leaves a free string of length $n-1$, while choosing 1 forces the next bit and leaves $n-2$ free, which is why the equation is $f(n) = f(n-1) + f(n-2)$. The boundary conditions supply the cases too small to decompose. So the whole setup is Lecture 31's addition and multiplication principles, applied to a problem whose subcases are versions of itself.

Full credit requires naming the case split as disjoint and exhaustive. An answer that just reports the equation earns about half.

Instructor notes

- **Problem 2 reaches Fibonacci from a completely different problem.** Worth putting beside the stair-climbing example — same recurrence, different boundary conditions, different sequence.
- **Problem 4** establishes the boundary-condition rule, which students get wrong constantly when they start solving on Wednesday.
- **Problem 3** is the one where the *reasoning* is hard and the algebra easy. Good to work at the board.
- **Problem 1(c)** keeps the Week 1 standard alive: terms are evidence.